



SCAMNET

Network-Level Fraud Detection for Online Marketplaces & Job Postings

Detect the network behind the scam not just the suspicious post.

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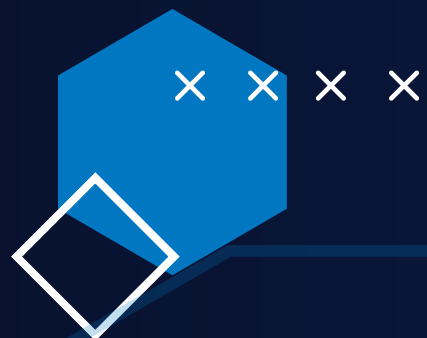
PROBLEM STATEMENT

Online fraud is becoming a network problem

- Fake job listings, marketplace scams and phishing posts can deceive users through seemingly legitimate listings.
- Existing defenses often rely on reports, keyword filters and individual-listing analysis.
- Coordinated scammers can operate through multiple accounts while reusing images, contact details and posting patterns.
- When each listing is viewed independently, these connections remain hidden.

The Gap

The evidence of fraud often isn't inside one post, it's in the relationships between many posts.



PROBLEM UNDERSTANDING

Why individual-level detection misses coordinated fraud

Listing A	Listing B	Listing C
iPhone 15 Pro	Apple iPhone — urgent sale	iPhone 15 Pro Max
₹52,000	₹49,999	₹54,000
📞 98XXXX21	📞 98XXXX21	📞 97XXXX84

Individually → they may look legitimate.

Together →

shared contact + reused image + synchronized activity

→ 🚨 Potential coordinated network

Core insight

A scam ring leaves a fingerprint across multiple listings.



PROPOSED SOLUTION

SCAMNET:

A fraud-detection platform that cross-references every new listing against existing listings and identifies suspicious relationships.

Three detection signals



Image Reuse

Custom perceptual dHash detects reused images even after resizing or compression.



Contact Reuse

Identifies the same phone number appearing across different listings/accounts.



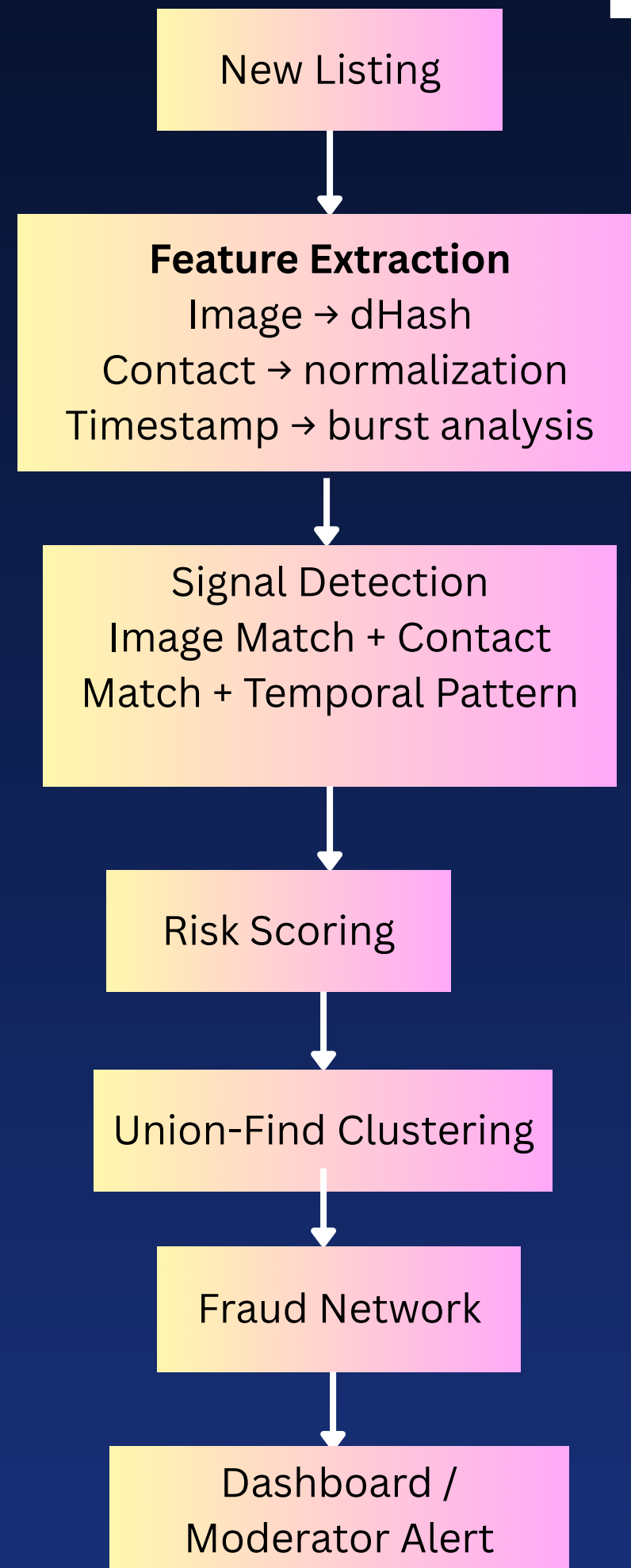
Temporal Bursts

Detects unusual concentrations of listings within short time windows.

These signals are combined to generate a **risk score** and connect related listings into fraud networks.

From “Is this listing suspicious?” → “Is this listing connected to a suspicious network?”

TECHNICAL WORKFLOW



1.0 HIGH RISK

iPhone 13 selling fast, DM now
Barely used, all accessories included, cash only
📍 Indore
💰 ₹27,000
Vikram Thakur · 9812345603



⚠️ Image matches: "iPhone 13 for sale, urgent", "iPhone 13 Pro cheap price, must sell today"

⚠️ Contact number reused under a different name

⚠️ Part of an unusual burst of 3 posts within 15 minutes

0.5 SUSPICIOUS

iPhone 13 Pro cheap price, must sell today
Excellent condition, selling due to upgrade, genuine buyers only
📍 Indore
💰 ₹26,500
Sneha Patil · 9812345604



⚠️ Image matches: "iPhone 13 for sale, urgent"

0.0 CLEAR

iPhone 13 for sale, urgent
Selling my iPhone 13, barely used, need cash fast, price negotiable
📍 Indore
💰 ₹28,000
Rohan Malhotra · 9812345603



0.0 CLEAR

Part-time content writer needed
Looking for a freelance writer for blog content, flexible hours, remote work
📍 Remote
💰 ₹8,000/month

0.0 CLEAR

Study table with drawer, good condition
Selling my wooden study table, minor scratches, pickup only
📍 Bhopal

SCAMNET IN ACTION

ScamNet

● LIVE FEED

* Light

Catch scam rings before they scale.

ScamNet cross-references images and contact details across every listing to surface coordinated fraud — not just single bad posts.

+ New Listing

View Fraud Networks

All

Marketplace

Jobs

5

TOTAL POSTS

2

FLAGGED

1

HIGH RISK

1

SUSPICIOUS NETWORKS

FRAUD NETWORK DETECTION

3 Listings
3 Accounts
Shared Contact
3 Reused Images
● High Risk

Suspicious Networks

AllMarketplaceJobsClose

Network #01

Risk: 50%

3 listings

3 accounts

shared contact



iPhone 13 for sale, ...
Rohan Malhotra



iPhone 13 Pro chea...
Sneha Patil



iPhone 13 selling fa...
Vikram Thakur

IMPACT

Making coordinated fraud harder to operate

🛡️ Earlier detection — surface suspicious networks before they spread.

💰 Reduce potential losses — warn users and platforms about connected high-risk activity.

🔍 Faster investigation — provide relationships and evidence instead of isolated alerts.

🌐 Platform-agnostic approach — the detection layer can be adapted to marketplaces and job platforms.

⚠️ Raise the cost of coordinated scams — reused assets and identifiers become detectable signals.

TECHNICAL IMPLEMENTATION

Layer	Technology
Frontend	React + Vite
Backend	Java + Spring Boot
Persistence	MySQL + Spring Data JPA
Image Detection	Custom perceptual dHash
Network Clustering	Union Find/ Disjoint Set
API	REST
Deployment	https://scamnet-vert.vercel.app/

Why this stack?

Fast, self-contained and explainable, no dependency on expensive external ML APIs.

WHY SCAMNET IS DIFFERENT

Beyond Rule-Based Scam Detection

Traditional Approach	SCAMNET
Analyze one listing	Analyze relationships between listings
Keyword / rule based	Multiple behavioral & content signals
Flag individual posts	Cluster connected activity
Reactive reports	Proactive pattern detection
“This looks suspicious”	“These listings form a suspicious network”

Innovation

SCAMNET treats fraud as a network problem rather than an individual-content problem.

FEASIBILITY & FUTURE SCOPE

Current MVP

Marketplace & job listings
Image upload + perceptual hashing
Contact reuse detection
Temporal burst detection
Risk scoring
Fraud network clustering
Interactive dashboard

Next

- Semantic similarity between listing descriptions
- Cross-platform ingestion
- More entity types: email, payment identifiers, device fingerprints
- Real-time moderator alerts
- Larger-scale graph analysis

THANK YOU

SCAMNET

Don't just detect the scam.
Detect the network behind it.

GitHub: <https://github.com/oshin-30/ScamNet>

Demo: <https://scamnet-vert.vercel.app/>

